

Math 161

Midterm Exam 1

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1. The logical operator “exclusive or,” often abbreviated “xor,” is defined by the following truth table:

P	T	T	F	F
Q	T	F	T	F
$P \text{ xor } Q$	F	T	T	F

Use a truth table to prove that for any propositions P , Q , the proposition “ $P \text{ xor } Q$ ” is logically equivalent to

$$((\text{not } P) \text{ and } Q) \text{ or } (P \text{ and } (\text{not } Q)).$$

(Make sure to show your work by including some intermediate steps, i.e. your truth table should have more than three rows.)

2. Prove that for any $n \in \mathbb{N}$, we have

$$1^2 + 3^2 + \cdots + (2n-1)^2 = \frac{4n^3 - n}{3}.$$

3. Let $A, B \subset \mathbb{R}$ be nonempty sets of real numbers such that $x \leq y$ for all $x \in A$ and $y \in B$. Prove that $\sup A$ and $\inf B$ exist, and that $\sup A \leq \inf B$.